

Module 1: Lesson 4

## Fourier Methods for Heston Model



# Outline

- ▶ Heston Pricing via Lewis (2001)
- ▶ Heston Model Calibration

# Heston Pricing via Lewis (2001)

In this lesson, we will apply the characteristic function pricing method by Lewis (2001) for option pricing in the Heston (1993) model.

Remember that our baseline pricing equation is Lewis (2001) integral value:

$$C_0 = S_0 - \frac{\sqrt{S_0 K} e^{-rT}}{\pi} \int_0^\infty \text{Re}[e^{izk} \varphi(z - i/2)] \frac{dz}{z^2 + 1/4}$$

Once we have this equation as a reference for pricing, we need at least 2 more things:

1. A characteristic function,  $\varphi(z - i/2)$ , for the Heston (1993) model.
2. A way to numerically calculate the integral value  $\rightarrow$  quadrature method.

In general, you can see that the problem will most often lie in unveiling the characteristic function for the process followed by the underlying asset. Let's then derive the Heston (1993) characteristic function.

# Heston (1993) characteristic function

Obtaining the characteristic function,  $\varphi^H(u, T)$ , for Heston will not be as straightforward as was the case for Black-Scholes. Please check the additional material to see the detailed derivation:

$$\varphi^H(u, T) = e^{H_1(u, T) + H_2(u, T)\nu_0}$$

$$H_1(u, T) \equiv r_0 uiT + \frac{c_1}{\sigma_\nu^2} \left\{ (\kappa_\nu - \rho\sigma_\nu ui + c_2)T - 2\log \left[ \frac{1 - c_3 e^{c_2 T}}{1 - c_3} \right] \right\}$$

$$H_2(u, T) \equiv \frac{\kappa_\nu - \rho\sigma_\nu ui + c_2}{\sigma_\nu^2} \left[ \frac{1 - e^{c_2 T}}{1 - c_3 e^{c_2 T}} \right]$$

$$c_1 \equiv \kappa_\nu \theta_\nu$$

$$c_2 \equiv -\sqrt{(\rho\sigma_\nu ui - \kappa_\nu)^2 - \sigma_\nu^2(-ui - u^2)}$$

$$c_3 \equiv \frac{\kappa_\nu - \rho\sigma_\nu ui + c_2}{\kappa_\nu - \rho\sigma_\nu ui - c_2}$$

# Heston (1993) calibration

As always, the previous characteristic function and resulting pricing equation has a bunch of parameters that we do not know ex-ante.

How do we obtain these parameters? → **Calibration!**

Specifically, there are 5 parameters in Heston (1993) that we need to calibrate:

$$\kappa_{\nu}, \theta_{\nu}, \sigma_{\nu}, \rho, \nu_0$$

Since now, under the Lewis (2001) approach, we do have a functional form for the price of a call option, we can calibrate the model to option market quotes and obtain these parameters.

We will do this, as always, via an optimization procedure that iterates over the different parameter values, aiming to minimize the error between model-obtained and market option prices. For faster convergence of the model, we will perform the calibration in 2 steps:

1. Focus on more sensible regions for the parameters ('Brute' force).
2. Within sensible regions, dig deeper to obtain accurate estimates.

# Summary of Lesson 4

In Lesson 4, we looked at:

- ▶ Price under the Heston (1993) model following the Lewis (2001) approach
- ▶ Using the functional form for Heston (1993) provided by Lewis (2001) to calibrate the model to option market quotes

⇒ **TO DO NEXT:** In the notebook associated with this lesson, you will find a detailed practical example of pricing and calibration under the Heston (1993) model.

⇒ In the next module, we will repeat the same pricing and calibration process for a more complex model that incorporates not only stochastic volatility, but also jump diffusion.